
CONTRASTIVE REPRESENTATION LEARNING FOR PROMPT-RESPONSE ALIGNMENT

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ABSTRACT

Reinforcement learning from human feedback (RLHF) has become a standard approach for aligning large language models with human preferences, but reward models may rely on superficial features such as length, style, or fluency rather than true adherence to prompt intention. This thesis studies prompt-response alignment as a representation learning problem and proposes TextCLIP, a contrastive framework that learns a shared embedding space for prompts and responses. Instead of measuring absolute response quality, TextCLIP learns a prompt-conditioned compatibility function that captures how well a response addresses the semantic goal and constraints of a given prompt.

TextCLIP uses dual encoders with projection heads and is trained with a symmetric contrastive objective that pulls matched prompt-response pairs together while pushing mismatched pairs apart. I evaluate both a lightweight TextCLIP model trained from scratch and a pretrained variant built on BAAI/bge-base-en-v1.5. On the hardest corresponding-rewrite task, the original TextCLIP achieves 0.768 accuracy, while the pretrained TextCLIP reaches 0.856, outperforming a pretrained embedding similarity baseline. Length-bias analysis further suggests that the model does not simply reward longer responses when added content becomes off-topic.

I also apply DPO-style fine-tuning to adapt the shared embedding space to on-topic, toxicity-related, and general preference tasks. Fine-tuning substantially improves pairwise accuracy and makes the separation between chosen and rejected responses more pronounced in UMAP visualizations. Layer-wise probing results show that alignment-related signals are already linearly decodable from TextCLIP’s internal representations, with different tasks relying on different depths of the encoder. Overall, these results suggest that contrastive representation learning provides a useful and interpretable framework for modeling prompt-specific alignment.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) has become a widely adopted framework for aligning large language models (LLMs) with human preferences (Christiano et al., 2017a). But it also introduces the risk of reward hacking. Instead of truly following user intent, models may learn to exploit weaknesses in the reward model to obtain higher scores. In particular, reward models often rely on superficial features such as response length, fluency, or style, leading to biases like length bias (Amodei et al., 2016). As a result, a response can receive a high reward even if it does not meaningfully address the prompt.

Prior work often tries to remove these specific spurious artifacts. However, features such as length are not always irrelevant. In many cases, they are part of the prompt intention itself—for example, when a user asks for a detailed explanation or a brief answer. This means the key problem is not simply to eliminate particular artifacts, but to measure how well a response addresses the intention of the prompt. More broadly, we need a function that captures whether a response is truly on-topic for the given prompt.

This thesis studies that question through a contrastive representation learning framework, referred to as TextCLIP, for modeling prompt–response alignment. Inspired by CLIP, which learns a shared embedding space for paired images and texts, TextCLIP aims to learn a joint embedding space in which matched prompts and responses are placed close together, while mismatched pairs are pushed apart. The underlying intuition is that contrastive learning can encourage the model to capture relational information: a good response is not defined in isolation, but by its compatibility with a particular prompt. By training on prompt–response pairings rather than only scalar preference labels, the model may be able to learn a representation that better reflects prompt-conditional semantic alignment.

The motivation for this approach is twofold. First, contrastive learning provides a natural way to model prompt-specific compatibility. For each prompt, matched responses can be treated as positive examples, while responses to other prompts serve as negatives. This shifts the learning problem away from fitting broad preference patterns and toward identifying which responses actually reflect the intention of a given prompt. Under this setup, surface features are not removed by assumption; instead, the model is encouraged to distinguish features that are genuinely related to the prompt from those that are merely incidental. Second, a shared embedding space offers an explicit representation of prompt–response alignment. Such a representation is useful not only for scoring, but also for retrieval, response ranking, and analyzing how alignment signals are organized geometrically. If the learned space captures prompt intention and response relevance, it can also help reveal when standard reward models rely on factors such as length or style for the wrong reasons.

The central hypothesis of this thesis is that contrastive learning can provide a more robust signal of prompt–response alignment than objectives that depend primarily on direct preference supervision. Specifically, I investigate whether a CLIP-style architecture can learn representations that distinguish semantically aligned responses from misaligned ones, while being less sensitive to superficial artifacts. This thesis therefore positions prompt–response alignment as a representation learning problem: the goal is not only to score outputs, but to construct an embedding space in which semantic compatibility between prompts and responses becomes geometrically meaningful.

To study this problem, I develop a TextCLIP framework in which prompts and responses are encoded separately and projected into a shared latent space. The model is trained using a contrastive objective that encourages matched prompt–response pairs to have similar representations. In settings where multiple responses are available for the same prompt, these responses provide a particularly useful training signal, as they allow the model to capture prompt-level semantic consistency while contrasting against responses generated for different prompts. Through this framework, I examine whether the learned representations can serve as an alternative basis for measuring “on-topicness,” and whether they reduce the influence of spurious features that commonly affect preference-based evaluation.

This thesis makes three main contributions. First, it formulates prompt–response alignment as a contrastive representation learning problem and proposes TextCLIP as a CLIP-inspired framework for learning shared prompt and response representations. Second, it empirically shows that contrastive learning can improve prompt-specific alignment, both in a lightweight model trained from scratch and in a pretrained encoder setting. Third, it provides diagnostic analyses of the learned representation space, including length-bias evaluation, DPO-style preference adaptation, UMAP visualization, and layer-wise probing. Together, these results suggest that TextCLIP offers an interpretable and flexible framework for studying alignment beyond scalar reward modeling.

2 PRELIMINARY

2.1 REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017b; Ziegler et al., 2019) has become as a key technique for integrating human preference signals into machine learning methods. In particular, RLHF has become a standard component in the post-training pipe line of foundation Large Language Models (LLMs), which serves to align the outputs of these models with human values such as helpfulness, harmlessness, and honesty (Ouyang et al., 2022; Bai et al., 2022). Notable examples include the revolutionary closed-source ChatGPT (Achiam et al., 2023), Claude (Anthropic, 2023), and Gemini (Team et al., 2023), as well as the powerful open-source mod-

els like Zephyr(Tunstall et al., 2023) and LLaMA-3(Meta, 2024). In particular, since the introduction of ChatGPT, RLHF has attracted significant interest in a diverse set of communities.

However, recent work has revealed that reward models are susceptible to reward hacking, where models exploit imperfections in the learned reward function rather than genuinely aligning with human intent (Amodei et al., 2016). Reward hacking can arise from unintentional and prompt-irrelevant human preferences (Wang et al., 2025).

Preference Data and Reward Modeling Let x denote a prompt and y denote a response. A preference dataset can be written as

$$\mathcal{D}_{\text{pref}} = \{(x_i, y_i^w, y_i^l)\}_{i=1}^n,$$

where y_i^w is the chosen response and y_i^l is the rejected response for the same prompt. This gives the prompt-conditional preference relation

$$y_i^w \succ y_i^l \mid x_i.$$

Such pairwise supervision avoids requiring absolute quality scores and instead asks annotators to compare two responses under the same prompt.

A common reward modeling objective is the Bradley–Terry loss (Bradley and Terry, 1952):

$$\mathcal{L}_{\text{RM}}(\phi) = -\mathbb{E}_{(x, y_w, y_l)} [\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))], \quad (1)$$

where r_ϕ is the reward model and $\sigma(\cdot)$ is the logistic sigmoid function.

Direct Preference Optimization Direct Preference Optimization (DPO) directly optimizes a policy on preference pairs, avoiding the separate reward-modeling and reinforcement-learning stages of standard RLHF (Zhao et al., 2023). Let $\pi_\theta(y \mid x)$ be the optimized policy and $\pi_{\text{ref}}(y \mid x)$ be a fixed reference policy. DPO encourages the policy to assign higher relative likelihood to the chosen response than to the rejected response:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y^w, y^l) \sim \mathcal{D}_{\text{pref}}} \left[\log \sigma \left(\beta \left[\log \frac{\pi_\theta(y^w \mid x)}{\pi_{\text{ref}}(y^w \mid x)} - \log \frac{\pi_\theta(y^l \mid x)}{\pi_{\text{ref}}(y^l \mid x)} \right] \right) \right],$$

where $\beta > 0$ controls the strength of preference optimization.

In this thesis, we do not use TextCLIP similarity as a reward to optimize an autoregressive language model. Instead, we apply a DPO-style tuning idea to the TextCLIP encoders, aiming to reshape the shared embedding space so that preferred responses are closer to their prompts than rejected responses.

2.2 CONTRASTIVE LEARNING

Contrastive learning is an instance-based representation learning method (Wu et al., 2018; Jaiswal et al., 2020). It learns an embedding space where similar examples are close and dissimilar examples are far apart.

Let u and v be two related views, encoded as

$$z_u = f(u), \quad z_v = g(v). \quad (2)$$

Their similarity is often measured by cosine similarity:

$$\text{sim}(z_u, z_v) = \frac{z_u^\top z_v}{\|z_u\| \|z_v\|}. \quad (3)$$

For an anchor u_i , with positive pair v_i^+ and negatives $\{v_j\}_{j \neq i}$, the InfoNCE loss is

$$\mathcal{L}_{\text{NCE}} = -\sum_{i=1}^N \log \frac{\exp(\text{sim}(z_{u_i}, z_{v_i^+})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_{u_i}, z_{v_j})/\tau)}, \quad (4)$$

where $\tau > 0$ is the temperature parameter. This objective increases similarity between positives while reducing relative similarity to negatives.

A useful view is that contrastive learning balances *alignment* and *uniformity*: alignment pulls positive pairs together, while uniformity prevents representation collapse by spreading embeddings across the space (Wang and Isola, 2020).

CLIP. Contrastive Language–Image Pre-training (CLIP) learns aligned image and text representations with a dual-encoder contrastive objective (Radford et al., 2021). Matched image–text pairs are pulled closer in a shared embedding space, while mismatched pairs are pushed apart. This provides a general framework for learning semantic alignment through contrastive representation learning.

3 METHODOLOGY

3.1 PROBLEM SETUP

Let

$$\mathcal{D} = \{(p_i, \mathcal{R}_i)\}_{i=1}^N$$

denote a dataset of prompts and responses, where p_i is the i -th prompt and

$$\mathcal{R}_i = \{r_{i1}, r_{i2}, \dots, r_{iK}\}$$

is the set of K responses generated for the same prompt.

The goal of TextCLIP is to learn a similarity function $s(p, r)$ that measures how well a response r addresses the intention of prompt p . Instead of evaluating response quality in isolation, TextCLIP focuses on prompt-conditioned alignment. This is important because features such as length, style, or formatting are not inherently good or bad; their relevance depends on the specific prompt.

3.2 OVERVIEW OF TEXTCLIP

TextCLIP is inspired by CLIP and adapts its dual-encoder framework to prompt–response pairs. As shown in Fig 1, the model contains:

- a **prompt encoder** $f_\theta(\cdot)$,
- a **response encoder** $g_\phi(\cdot)$.

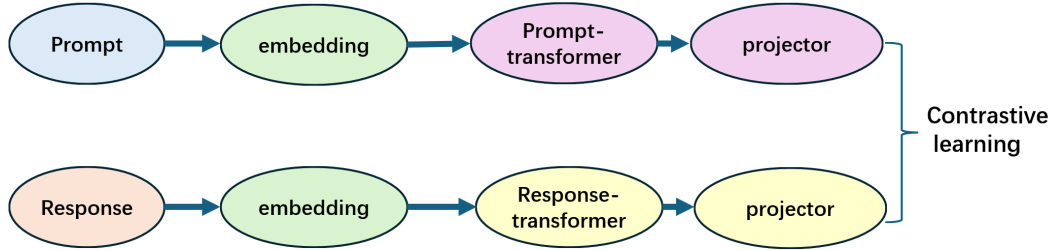


Figure 1: The framework of TextCLIP.

Given a prompt p and a response r , the encoders map them into a shared embedding space:

$$\mathbf{h}_p = f_\theta(p), \quad \mathbf{h}_r = g_\phi(r).$$

Matched prompt–response pairs are encouraged to have high similarity, while mismatched pairs are pushed apart.

3.3 ENCODER ARCHITECTURE

Each prompt and response is tokenized, embedded, and passed through a Transformer-based encoder. TextCLIP uses a dual-encoder structure:

$$\mathbf{z}_p = \text{Encoder}_p(p), \quad \mathbf{z}_r = \text{Encoder}_r(r),$$

where the prompt and response encoders have similar structures but do not share parameters.

The intermediate representations are projected into a shared embedding space:

$$\mathbf{h}_p = W_p \mathbf{z}_p + b_p, \quad \mathbf{h}_r = W_r \mathbf{z}_r + b_r.$$

The projected embeddings are then normalized:

$$\tilde{\mathbf{h}}_p = \frac{\mathbf{h}_p}{\|\mathbf{h}_p\|_2}, \quad \tilde{\mathbf{h}}_r = \frac{\mathbf{h}_r}{\|\mathbf{h}_r\|_2}.$$

For simplicity, we use $\mathbf{h}_p$ and $\mathbf{h}_r$ to denote the normalized embeddings.

3.4 SIMILARITY FUNCTION

Given a prompt embedding $\mathbf{h}_{p_i}$ and a response embedding $\mathbf{h}_{r_{kj}}$, their similarity is defined as scaled cosine similarity:

$$s(p_i, r_{kj}) = \frac{\mathbf{h}_{p_i}^\top \mathbf{h}_{r_{kj}}}{\tau},$$

where $\tau > 0$ is a temperature parameter.

This score measures prompt-specific compatibility rather than global response quality.

3.5 CONTRASTIVE LEARNING OBJECTIVE

For each prompt p_i , its associated responses

$$\{r_{i1}, r_{i2}, \dots, r_{iK}\}$$

are treated as positive examples. Responses from other prompts are treated as negatives.

For each matched pair (p_i, r_{ij}) , the InfoNCE-style loss is

$$\mathcal{L}_{i,j} = -\log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \sum_{m=1}^K \exp(s(p_i, r_{km}))}.$$

The full objective is

$$\mathcal{L}_{\text{TextCLIP}} = -\frac{1}{NK} \sum_{i=1}^N \sum_{j=1}^K \log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \sum_{m=1}^K \exp(s(p_i, r_{km}))}.$$

This objective pulls each prompt closer to its matched responses and pushes it away from responses associated with other prompts.

Multi-positive Symmetric Loss. In practice, TextCLIP uses a multi-positive symmetric contrastive loss. Let $\mathcal{R}_i^+$ denote the matched responses for prompt p_i , and let

$$\mathcal{R} = \bigcup_{i=1}^N \mathcal{R}_i^+$$

be all responses in the minibatch. The similarity score is

$$s(p_i, r) = \alpha z_{p_i}^\top z_r,$$

where α is a learnable temperature parameter.

The prompt-to-response loss is

$$\mathcal{L}_{p \rightarrow r} = \frac{1}{N} \sum_{i=1}^N \left[- \sum_{r \in \mathcal{R}_i^+} \log \frac{\exp(s(p_i, r))}{\sum_{r \in \mathcal{R}} \exp(s(p_i, r))} \right].$$

The response-to-prompt loss is

$$\mathcal{L}_{r \rightarrow p} = \frac{1}{NK} \sum_{i=1}^N \sum_{j=1}^K \left[- \log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \sum_{m=1}^K \exp(s(p_k, r_{im}))} \right].$$

The final objective is

$$\mathcal{L}_{\text{TextCLIP}} = \frac{1}{2} (\mathcal{L}_{p \rightarrow r} + \mathcal{L}_{r \rightarrow p}).$$

This symmetric objective encourages both prompt-to-response and response-to-prompt alignment. When each prompt has only one matched response, it reduces to the standard CLIP-style contrastive loss.

Optimization. The model is trained by minimizing

$$\min_{\theta, \phi} \mathcal{L}_{\text{TextCLIP}}.$$

The encoder parameters and temperature parameter are optimized jointly using Adam-style optimization.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

We evaluate TextCLIP on several prompt–response alignment tasks. The experiments are designed to answer three questions: whether TextCLIP can identify responses that better match the prompt intention, whether its score is robust to superficial length variation, and whether the learned embedding space can be further adapted to different preference objectives.

We use two types of data. First, for contrastive training, we construct a 20K prompt–response dataset from SmolTalk (allal et al., 2025) and AlpacaFarm (Dubois et al., 2023). For each prompt, we augment the original response with completions generated by DeepSeek-V3.1-Base (DeepSeek-AI et al., 2025), LLaMA3-72B (Grattafiori et al., 2024), and Qwen3-235B-A22B (Yang et al., 2025). Thus, each prompt is associated with multiple responses. This construction allows TextCLIP to learn prompt-level compatibility from responses that share the same user intention but may differ in wording, style, and length.

Second, for evaluation and DPO-style fine-tuning, we use prompt–chosen–rejected triples. For on-topic evaluation, we construct pairwise comparisons from a subset of RewardBench2. In addition to the original chosen and rejected responses, we use several rewrite-based settings to test whether the model captures prompt-specific relevance rather than only broad semantic similarity. We also evaluate on toxicity-related preference data from Toxic-DPO v0.2 (unalignment, 2024) and general human preference data from a binarized version of UltraFeedback (Cui et al., 2024).

We compare two TextCLIP variants. The first is Original TextCLIP, a lightweight dual-encoder model trained from scratch. The second is Pretrained TextCLIP, which uses BAAI/bge-base-en-v1.5 as the pretrained backbone and adds separate prompt and response projection heads. Both models encode prompts and responses separately, map them into a shared embedding space, and compute compatibility using normalized dot-product similarity.

At test time, given a prompt p , a chosen response r^+ , and a rejected response r^- , the model predicts correctly if

$$s(p, r^+) > s(p, r^-).$$

We report pairwise accuracy as the main metric. This evaluation protocol is used across on-topic response selection, toxicity-related preference, and general human preference tasks.

4.2 ON-TOPIC RESPONSE SELECTION

We first evaluate whether TextCLIP can measure how well a response addresses the prompt intention. We compare TextCLIP with 3 baselines: the OpenAssistant reward model OpenAssistant/reward-model-deberta-v3-large-v2, the Skywork reward model Skywork/Skywork-Reward-V2-Qwen3-8B, and the pretrained bge-base embedding similarity baseline. The evaluation includes three settings. The `corr-rewrite` setting is the hardest because the negative response is a corresponding rewrite related to the same prompt. The `random-rewrite` and `random-chosen` settings are easier because the negative response is sampled from another prompt.

Table 1: Accuracy on the on-topic response selection task.

Method	corr-rewrite	random-rewrite	random-chosen
OpenAssistant RM	0.612	0.764	0.768
Skywork RM	0.692	0.852	0.776
Original TextCLIP	0.768	0.856	0.917
bge-base Embedding Similarity	0.812	1.000	0.988
Pretrained TextCLIP	0.856	1.000	0.993

Table 1 shows that TextCLIP substantially improves over the reward-model baseline. The OpenAssistant reward model and Skywork reward model both perform weakly across all settings, especially on the harder `corr-rewrite` task. This suggests that a general reward model does not necessarily capture fine-grained prompt-response correspondence.

Original TextCLIP achieves 0.768 accuracy on the `corr-rewrite` setting, even though it is trained from scratch with a relatively lightweight encoder. This indicates that the contrastive learning framework itself can learn useful prompt-response compatibility signals. The pretrained bge-base embedding baseline is stronger, achieving 0.812 accuracy on `corr-rewrite` and near-perfect performance on the easier random-negative settings. However, Pretrained TextCLIP further improves `corr-rewrite` accuracy to 0.856 while maintaining near-perfect accuracy on the random-negative settings. These results suggest that contrastive fine-tuning can adapt pretrained semantic embeddings toward prompt-conditioned alignment.

4.3 LENGTH-BIAS EVALUATION

A central motivation of TextCLIP is to avoid relying on superficial features such as response length. To test this, we evaluate whether longer responses receive higher compatibility scores simply because they are longer. We consider two settings. In the first, the longer rewrite preserves the content and topic of the original response. In the second, the longer rewrite introduces off-topic content while remaining fluent and polite.

We report the average treatment effect (ATE) to quantify the effect of increasing response length on the TextCLIP compatibility score. For each prompt-response pair, we compare the score of the treated response, i.e., the longer rewrite, with the score of the corresponding control response, i.e., the shorter original or shorter rewrite. Formally, the ATE is computed as the average score difference between the longer and shorter responses:

$$\text{ATE} = \mathbb{E}[s(p, r)|\text{response}=\text{long}] - \mathbb{E}[s(p, r)|\text{response}=\text{short}] = \mathbb{E}[s(p, r_{\text{long}}) - s(p, r_{\text{short}})],$$

where $s(p, r)$ denotes the TextCLIP compatibility score between prompt p and response r . A positive ATE would indicate that longer responses tend to receive higher scores, suggesting a possible length bias. In contrast, an ATE close to zero suggests that length alone does not systematically affect the score, while a negative ATE indicates that the longer response is scored lower on average. Therefore, the ATE provides a direct way to test whether TextCLIP rewards verbosity or instead remains sensitive to prompt-response relevance.

Table 2: Length-bias ATE evaluation.

Setting	ATE	95% CI
Shorter vs. longer rewrite	−0.019	[−0.040, 0.011]
Shorter vs. longer off-topic rewrite	−0.140	[−0.164, −0.113]

Table 2 shows that TextCLIP does not simply favor longer responses. When the longer rewrite remains on-topic, the estimated ATE is close to zero and the confidence interval includes zero. This indicates that increasing length alone does not significantly increase the compatibility score.

In contrast, when the longer rewrite introduces off-topic content, the ATE becomes strongly negative. The longer off-topic rewrite receives a substantially lower compatibility score than the shorter

on-topic response. This suggests that TextCLIP is sensitive to prompt-specific relevance: additional length is not rewarded when it weakens the connection between the response and the prompt. Overall, the length-bias results support the claim that TextCLIP measures prompt-conditioned compatibility rather than superficial verbosity.

4.4 DPO-STYLE FINE-TUNING

We next examine whether the TextCLIP embedding space can be adapted to different alignment-related objectives. Starting from the contrastively trained model, we apply DPO-style fine-tuning on three pairwise tasks: on-topic response selection, toxicity-related preference, and general human preference. In each task, the data are organized as prompt–chosen–rejected triples, and the model is optimized to assign a higher compatibility score to the chosen response than to the rejected response.

Table 3: Pairwise accuracy before and after DPO-style fine-tuning.

Task	Training Stage	DPO Stage
On-topic response selection	76.9%	94.8%
Toxicity-related preference	80.4%	99.6%
Human preference	61.7%	87.7%

Table 3 shows that DPO-style fine-tuning substantially improves pairwise accuracy across all three tasks. For on-topic response selection, accuracy increases from 76.9% to 94.8%, suggesting that preference optimization further strengthens the model’s ability to identify responses that better address the prompt. For toxicity-related preference, accuracy increases from 80.4% to 99.6%, indicating that the TextCLIP score can be adapted to safety-related distinctions. For general human preference, accuracy improves from 61.7% to 87.7%, showing that the framework is not limited to topical relevance and can also capture broader preference signals.

These results suggest that the shared embedding space learned by contrastive training is not fixed. Instead, it can be further reshaped by preference optimization. In this sense, TextCLIP can serve not only as a prompt–response compatibility model, but also as a flexible representation-learning framework for different alignment objectives.

4.5 EMBEDDING SPACE ANALYSIS

To better understand the learned representations, we visualize the shared embedding space using UMAP. Figure 2 compares the pretrained `bge-base` embedding model with Original TextCLIP on the on-topic response selection task.

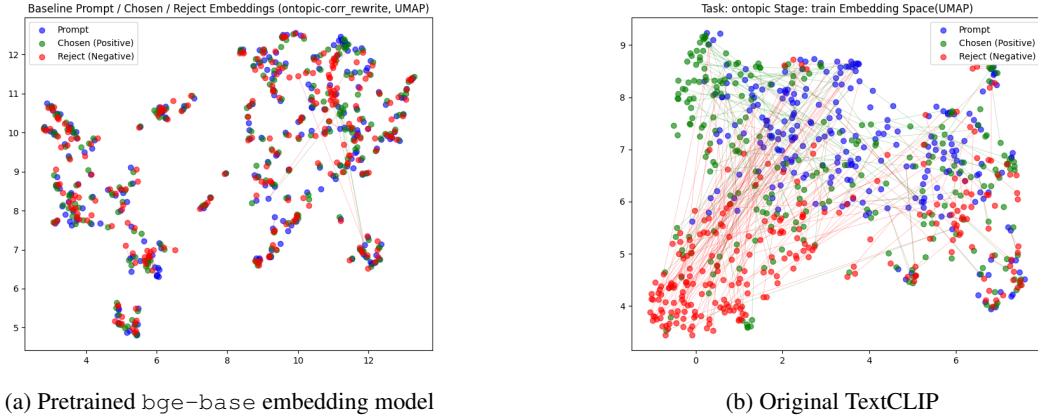


Figure 2: UMAP visualization of the shared prompt–response embedding space on the on-topic response selection task. Blue points denote prompts, green points denote chosen responses, and red points denote rejected responses.

In the pretrained `bge-base` embedding space, chosen and rejected responses are largely mixed. This suggests that off-the-shelf semantic embeddings capture general semantic relatedness, but do not clearly separate on-topic and off-topic responses. In contrast, Original TextCLIP produces a more structured embedding space: chosen responses tend to stay closer to the prompt region, while rejected responses form a more distinct cluster. This indicates that contrastive training helps TextCLIP learn representations that are more sensitive to prompt–response compatibility.

We further examine how DPO-style fine-tuning changes the embedding geometry. As shown in Appendix B.4, the separation between chosen and rejected responses becomes more pronounced after fine-tuning across on-topic, toxicity-related, and human-preference tasks. This qualitative pattern is consistent with the accuracy improvements in Table 3, suggesting that DPO-style fine-tuning not only improves pairwise prediction accuracy, but also reshapes the shared embedding space.

Overall, the UMAP results provide qualitative support for the quantitative findings. Contrastive training helps TextCLIP learn a prompt-sensitive embedding space, and DPO-style fine-tuning further reshapes this space by pulling chosen responses closer to prompts and pushing rejected responses farther away.

5 CONCLUSIONS

This paper proposes TextCLIP, a contrastive representation learning framework for modeling prompt–response alignment. Rather than treating response quality as an absolute property, TextCLIP learns a prompt-conditioned compatibility function that measures whether a response addresses the intention, constraints, and semantic goal of a given prompt.

The experiments show that TextCLIP can learn a meaningful shared embedding space for prompts and responses. On the on-topic corresponding-rewrite task, the original lightweight TextCLIP achieves strong performance despite being trained from scratch, while the pretrained TextCLIP further improves accuracy and outperforms the pretrained embedding similarity baseline. These results suggest that contrastive learning can adapt both lightweight and pretrained representations toward prompt-specific alignment.

The length-bias analysis further indicates that TextCLIP does not simply reward longer responses. When longer rewrites preserve the original content, the average score difference is close to zero; when longer rewrites introduce off-topic content, they receive substantially lower compatibility scores. This supports the claim that TextCLIP is sensitive to semantic fit rather than only surface-level length.

DPO-style fine-tuning improves pairwise accuracy across on-topic, toxicity-related, and general preference tasks. The UMAP visualizations show that fine-tuning also reshapes the shared embedding space by making chosen and rejected responses more separable. Overall, these findings suggest that TextCLIP provides a useful and interpretable framework for studying prompt-specific alignment, with potential applications in response ranking, reward-model diagnostics, and future policy optimization.

Finally, layer-wise probing provides a complementary diagnostic view of how alignment-related information is encoded inside the model. The probing results show that on-topic and toxicity-related signals are strongly linearly decodable from TextCLIP’s internal representations, suggesting that these forms of alignment are explicitly represented in the residual stream. In contrast, general human preference is more complex and less linearly separable before fine-tuning, likely because it depends on multiple factors such as relevance, helpfulness, specificity, and style. After DPO fine-tuning, preference-related information becomes more accessible in deeper layers, indicating that fine-tuning helps the model encode more abstract preference signals.

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A REWRITING INSTRUCTION

We describe the rewriting strategy to construct the evaluation data sets for on-topic task in Table 4 followed by a specific example of the rewriting results A.

Rewriting Type	Longer Rewrite	Shorter Rewrite	Longer Off-topic Rewrite
Rewriting Instruction	Rewrite the response to be longer while preserving meaning and style.	Make the previous response more concise without losing core information.	Expand the response with stylistic elaboration, allowing mild topical drift.

Table 4: Rewriting instructions used to construct response variants for robustness testing. Each instruction targets different attributes such as response length or topicality.

Rewriting Example

Prompt: *Why are the Kardashians so famous?*

Longer Rewrite: The Kardashian family has undeniably emerged as a prominent and influential cultural phenomenon within contemporary American society. They are most notably represented by the reality television show *Keeping Up with the Kardashians*, which has garnered widespread critical acclaim and achieved record-breaking ratings throughout its various seasons and spin-offs. This show has not only captivated audiences but has also significantly shaped popular culture and social dynamics. Beyond their television presence, the Kardashian family has successfully ventured into various other fields, with multiple members pursuing thriving careers in fashion, entertainment, social media, and other diverse arenas. Their entrepreneurial spirit and ability to capitalize on their fame have led to the creation of numerous brands and products that resonate with fans and consumers alike. Overall, the Kardashian family serves as a striking representation of the increasing prevalence of celebrity and media culture in contemporary American society. Their influence underscores the importance of images and narratives in modern cultural discourse, illustrating how celebrity status can shape public perception and societal trends.

Shorter Rewrite: The Kardashian family has become a significant cultural phenomenon in contemporary American society, primarily through their reality show *Keeping Up with the Kardashians*, which has received critical acclaim and achieved record-breaking ratings. This show has not only captivated audiences but also shaped popular culture and social dynamics. Beyond television, the family has successfully ventured into various fields, with members building thriving careers in fashion, entertainment, and social media. Their entrepreneurial spirit has led to numerous brands and products that resonate with fans and consumers. Overall, the Kardashian family exemplifies the rise of celebrity and media culture in modern America, highlighting how celebrity status can influence public perception and societal trends.

Longer Off-topic Rewrite: The Kardashian family has undeniably emerged as a prominent and influential cultural phenomenon within contemporary American society, captivating audiences with their unique blend of glamour, drama, and entrepreneurial spirit. Most notably represented by the reality television show *Keeping Up with the Kardashians*, which has garnered widespread critical acclaim and achieved record-breaking ratings across its various

seasons, the family’s impact extends far beyond the confines of the small screen. Each member of the family has carved out a niche for themselves, pursuing successful careers in diverse fields such as fashion, entertainment, and social media, thereby illustrating the multifaceted nature of their influence. This phenomenon raises intriguing questions about the evolving landscape of celebrity culture and the ways in which images and narratives shape public perception. Moreover, one might consider how the rise of social media platforms has transformed the way we engage with celebrities, blurring the lines between public and private life, and fostering a culture of constant connectivity. It’s fascinating to think about how this shift has not only affected the Kardashians but also countless other public figures who navigate the complexities of fame in the digital age. Overall, the Kardashian family’s prominence serves as a reflection of the increasing prevalence of celebrity and media culture in contemporary American society, highlighting the significance of visual storytelling and the narratives we construct around public personas. What does this say about our collective values and the way we consume media?

A ADDITIONAL EXPERIMENTAL DETAILS

A.1 DATASET CONSTRUCTION

We use two types of data: prompt–response data for contrastive training and prompt–chosen–rejected triples for pairwise evaluation and DPO-style fine-tuning.

For contrastive training, each prompt p_i is associated with multiple responses:

$$\mathcal{R}_i = \{r_{i1}, r_{i2}, \dots, r_{iK}\}.$$

Responses associated with the same prompt are treated as positives, while responses from other prompts are treated as negatives.

For on-topic evaluation, we construct several test modes:

- **corr-rewrite**: the negative response is a corresponding rewrite from the same prompt;
- **random-rewrite**: the negative response is a rewrite sampled from another prompt;
- **random-chosen**: the negative response is a chosen response sampled from another prompt;
- **length-bias**: the model compares shorter and longer rewrite variants.

We also evaluate on two auxiliary alignment datasets. The toxicity-related task uses Toxic-DPO v0.2 (unalignment, 2024), and the human-preference task uses a binarized version of UltraFeedback (Cui et al., 2024).

A.2 MODEL DETAILS

Original TextCLIP. Original TextCLIP is a lightweight dual-encoder model trained from scratch. It uses separate Transformer encoders for prompts and responses. The maximum context length is 1024, the batch size is 32, the vocabulary size is 512, and the Transformer width is 256. Each encoder contains 4 Transformer layers with 8 attention heads.

After encoding, mean pooling is applied over the sequence dimension:

$$\bar{h}(x) = \frac{1}{T} \sum_{t=1}^T h_t.$$

The pooled representation is passed through a linear projection head and normalized:

$$z_p = \frac{g_p(\bar{h}_p)}{\|g_p(\bar{h}_p)\|_2}, \quad z_r = \frac{g_r(\bar{h}_r)}{\|g_r(\bar{h}_r)\|_2}.$$

The compatibility score is

$$s(p, r) = \alpha z_p^\top z_r,$$

where $\alpha = \exp(\gamma)$ is a learnable logit scale.

Pretrained TextCLIP. Pretrained TextCLIP replaces the lightweight Transformer encoders with BAAI/bge-base-en-v1.5. The pretrained encoder is shared by the prompt and response branches, while the two branches use separate two-layer MLP projection heads.

Token representations are aggregated using attention-mask-aware mean pooling:

$$\bar{h}(x) = \frac{1}{\sum_{t=1}^T m_t} \sum_{t=1}^T m_t h_t,$$

where m_t is the attention mask. The projector maps the encoder hidden representation into a 512-dimensional latent space:

$$g(\bar{h}) = W_2 \text{GELU}(W_1 \bar{h} + b_1) + b_2.$$

The final embeddings are normalized before similarity computation.

A.3 TRAINING DETAILS

Original TextCLIP is trained on `merged_dataset_20K.jsonl` with AdamW and a learning rate of 1×10^{-4} . The dataset is split into 90% training and 10% validation using random seed 42. Training is run for up to 100 epochs with early stopping.

We consider both single-positive and multi-positive training. In the single-positive setting, each prompt is paired with one response. In the multi-positive setting, all responses associated with the same prompt are treated as positives. The response tensor has shape $B \times K \times T$, where B is the batch size, K is the number of responses per prompt, and T is the sequence length.

Pretrained TextCLIP uses the same contrastive training framework, but initializes the encoder from BAAI/bge-base-en-v1.5. The logit scale is initialized as $\log(1/0.07)$ and clamped so that the effective scale does not exceed 100.

B ADDITIONAL RESULTS

B.1 TRAINING CURVES

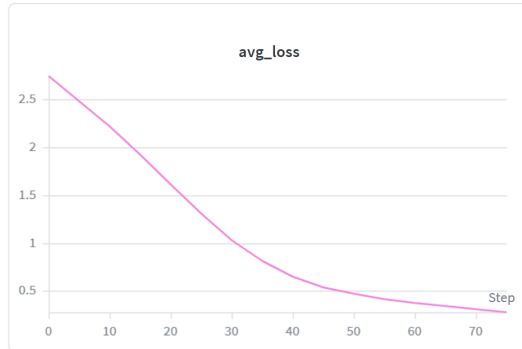


Figure 3: Training loss of Original TextCLIP.

B.2 SIMILARITY SCORE DISTRIBUTIONS

B.3 LENGTH-BIAS CORRELATION ANALYSIS

Table 5: Length-bias correlation evaluation.

Setting	Pooled Pearson	Pooled Spearman	Δ Pearson	Δ Spearman
Shorter vs. longer rewrite	0.149 ($p < 0.001$)	0.133 ($p = 0.003$)	-0.038 ($p = 0.547$)	-0.006 ($p = 0.924$)
Shorter vs. longer off-topic rewrite	-0.026 ($p = 0.561$)	-0.037 ($p = 0.411$)	-0.109 ($p = 0.086$)	-0.074 ($p = 0.244$)



Figure 4: Validation accuracy across training epochs.

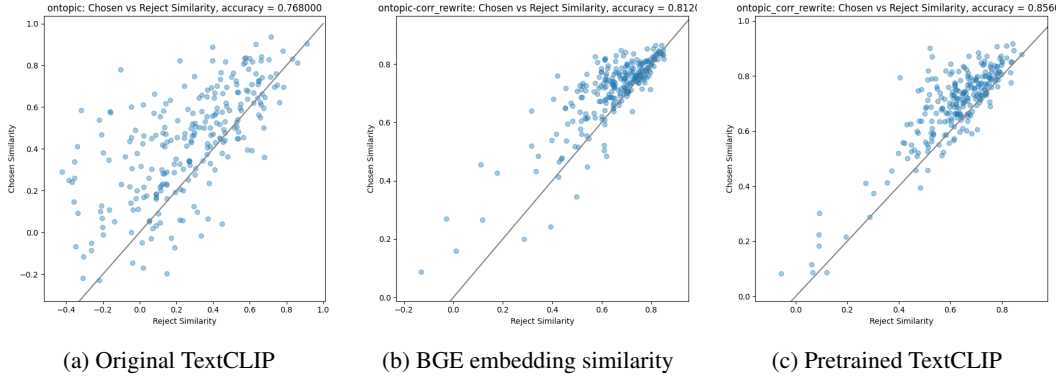


Figure 5: Similarity plots for the on-topic corresponding rewrite task.

The pooled correlation between length and score is weakly positive in the on-topic rewrite setting, but the prompt-controlled $\Delta\text{length}-\Delta\text{score}$ correlation is not significant. This suggests that the pooled correlation may reflect prompt-level variation rather than a direct preference for longer responses.

B.4 UMAP VISUALIZATIONS

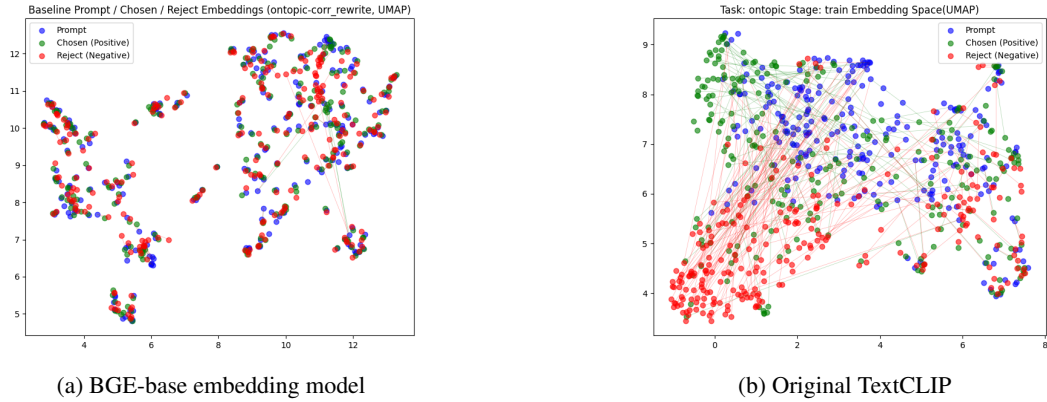


Figure 6: UMAP comparison between the BGE-base embedding model and Original TextCLIP.

C LAYER-WISE PROBING ANALYSIS

We conduct layer-wise probing to analyze whether alignment-related information is linearly decodable from TextCLIP’s internal representations. The encoder is frozen, residual representations

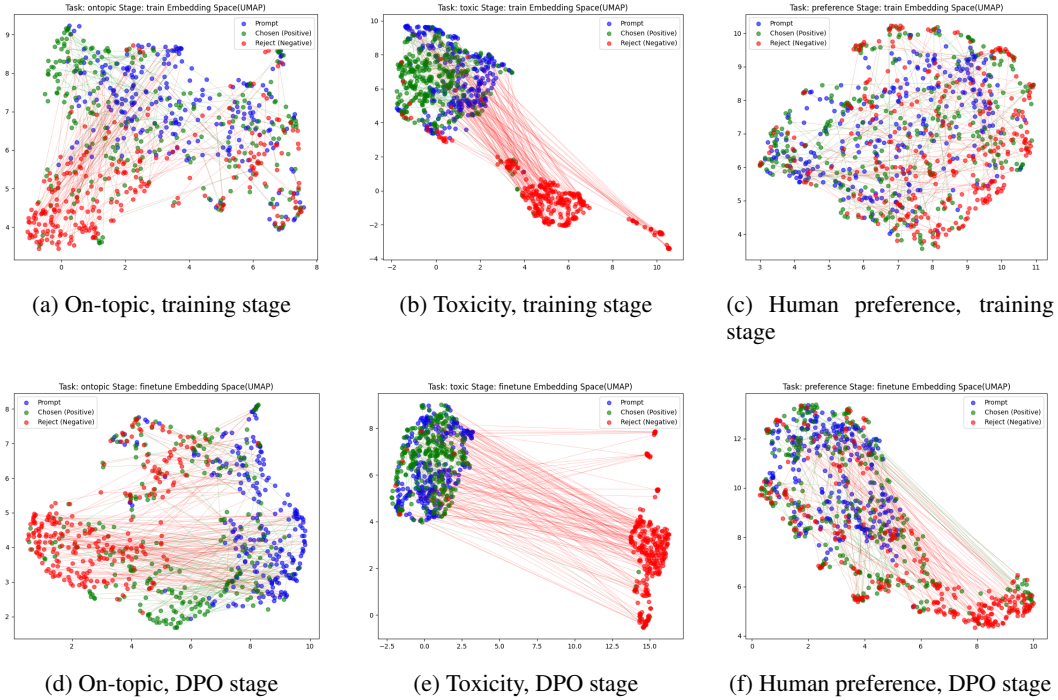


Figure 7: UMAP visualizations before and after DPO-style fine-tuning. Blue points denote prompts, green points denote chosen responses, and red points denote rejected responses.

are extracted from each Transformer layer, and a linear classifier is trained to distinguish chosen responses from rejected responses.

C.1 ON-TOPIC TASK

Table 6: Layer-wise probing results on the on-topic task.

2*Layer	Training Stage			DPO Stage		
	Accuracy	F1	AUC	Accuracy	F1	AUC
Layer 1	0.8700	0.8713	0.9484	0.8800	0.8800	0.9536
Layer 2	0.8600	0.8627	0.9472	0.8800	0.8824	0.9612
Layer 3	0.8400	0.8431	0.9332	0.8500	0.8544	0.9512
Layer 4	0.8400	0.8462	0.9196	0.8700	0.8738	0.9428

On-topic compatibility is strongly linearly decodable from internal representations. DPO-style fine-tuning improves performance across layers, suggesting that it strengthens alignment-relevant information throughout the encoder.

C.2 TOXICITY TASK

Toxicity-related preference signals are highly separable across all layers. DPO-style fine-tuning further improves accuracy, especially in the middle and deeper layers.

Table 7: Layer-wise probing results on the toxicity task.

2*Layer	Training Stage			DPO Stage		
	Accuracy	F1	AUC	Accuracy	F1	AUC
Layer 1	0.9724	0.9722	0.9919	0.9770	0.9770	0.9911
Layer 2	0.9770	0.9767	0.9905	0.9862	0.9862	0.9890
Layer 3	0.9770	0.9767	0.9909	0.9862	0.9862	0.9880
Layer 4	0.9770	0.9767	0.9915	0.9862	0.9862	0.9883

Table 8: Layer-wise probing results on the human preference task.

2*Layer	Training Stage			DPO Stage		
	Accuracy	F1	AUC	Accuracy	F1	AUC
Layer 1	0.5531	0.5699	0.5843	0.5475	0.5500	0.6117
Layer 2	0.5782	0.5770	0.6000	0.6089	0.6000	0.6617
Layer 3	0.5503	0.5360	0.5764	0.6034	0.5989	0.6391
Layer 4	0.5084	0.5111	0.5299	0.6285	0.6232	0.6578

C.3 HUMAN PREFERENCE TASK

Human preference is less linearly separable than on-topicness or toxicity. However, DPO-style fine-tuning improves deeper-layer probing performance, suggesting that general preference requires more abstract representations.